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CLINICAL RESEARCH
Digital health and innovation

Artificial intelligence–based quantification of breast arterial calcifications to predict cardiovascular morbidity and mortality

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Abstract

Background and Aims Women are underdiagnosed and undertreated for cardiovascular disease (CVD). Automatic quantification of breast arterial calcification (BAC) on screening mammography can identify women at risk for CVD. This study aimed to determine whether artificial intelligence–based automatic quantification of BAC from screening mammograms predicts CVD and mortality beyond PREVENT scores in a large, racially diverse, multi-institutional population.

Methods This retrospective cohort study included 123 762 women from two healthcare systems who had screening mammograms. Breast arterial calcification was quantified using a transformer-based neural network for segmentation. Breast arterial calcification severity was categorized as zero (0 mm^2), mild (>0–10 mm^2), moderate (>10–25 mm^2), and severe (>25 mm^2). Kaplan–Meier analysis, Cox proportional hazards, and Fine–Gray competing event models were used to examine the association between BAC and major adverse cardiovascular events (MACE).

Results Breast arterial calcification was detected in 16.1% (internal cohort) and 20.6% (external cohort) of women and provided significant prognostic value incremental to the PREVENT score. In PREVENT adjusted models, a clear dose–response was observed. Compared with zero BAC, mild [internal: hazard ratio (HR) 1.32, 95% confidence interval (CI) 1.10–1.59; external: HR 1.28, 95% CI 1.17–1.39], moderate (internal: HR 1.75, 95% CI 1.23–2.50; external: HR 1.79, 95% CI 1.55–2.06), and severe BAC (internal: HR 3.29, 95% CI 2.15–5.05; external: HR 2.80, 95% CI 2.36–3.32) were all prognostic for any MACE. Each 1 mm^2 increase in BAC conferred an additional 2%–3% risk for MACE ($P < .001$).

Conclusions Automatically quantified BAC is an independent predictor of MACE and mortality, adding prognostic value to the PREVENT score. This approach may provide an opportunistic cardiovascular risk assessment during routine mammography screening without additional radiation exposure to guide earlier and more effective preventive care for women.

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Structured Graphical Abstract

Key Question

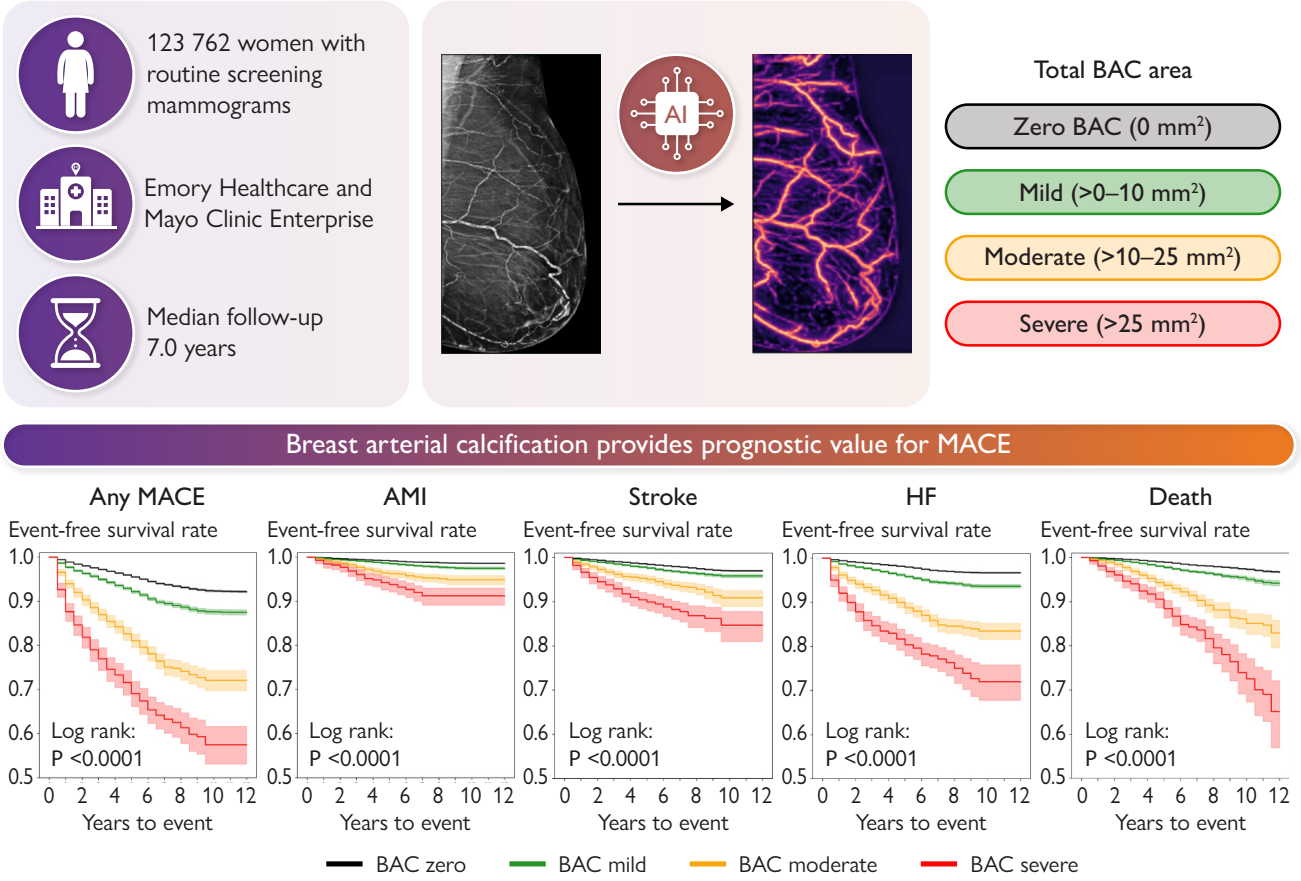
Is artificial intelligence-based automatic quantification of breast arterial calcification (BAC) on routine screening mammograms associated with cardiovascular outcomes?

Key Finding

In this multi-institutional cohort study of 123 762 women, mild, moderate, and severe BAC was associated with increased cardiovascular events compared to zero BAC. These associations remained significant across all age groups, including women under 50 years, and were independent of traditional risk factors and PREVENT risk score.

Take Home Message

These findings indicate an opportunity for secondary use of mammograms for early risk stratification during routine breast cancer screening.



Among 123 762 women undergoing screening mammography at two institutions, AI-based automated quantification of breast arterial calcification (BAC) demonstrated a continuous, dose-dependent association with major adverse cardiovascular events (MACE). Stratification from zero (0 mm^2) to severe BAC (>25 mm^2) was associated with a stepwise decline in event-free survival for all MACE including acute myocardial infarction, stroke, heart failure, and all-cause mortality across both sites (Log-rank $P < 0.0001$).

Keywords Breast arterial calcification (BAC) • Cardiovascular disease • Artificial intelligence • Mammography • Opportunistic screening • Risk prediction

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Introduction

Cardiovascular disease (CVD) is the leading cause of death in women yet is frequently underdiagnosed.[1–3] Risk assessment for CVD is typically performed using a combination of laboratory markers, family history, and other risk factors that are combined in various risk prediction models, with the PREVENT calculator currently recommended by guidelines.[4–6] However, a key limitation of these models is their reliance on clinical and laboratory data alone as they do not include any anatomic assessment of the vasculature.

Because of its high resolution and routine use in breast cancer screening, mammography provides an opportunity to directly visualize the vasculature in almost all adult women. Arterial calcification is readily apparent on screening mammograms, correlates with calcification in other arterial beds,[7] and can predict CVD events in the general population.[8,9] Unlike coronary artery calcification (CAC), which primarily reflects intimal-layer atherosclerotic changes leading to vessel narrowing, breast arterial calcification (BAC) occurs in the medial layer, resulting in increased arterial stiffness and decreased pulsatility rather than luminal obstruction.[7,10] Although BAC shares certain risk factors with atherosclerosis (such as age and diabetes), it also independently predicts CVD beyond these factors.[11,12] Thus, medial arterial calcification represents a distinct contribution to CVD risk, suggesting that incorporating BAC evaluation can enhance overall cardiovascular risk assessment beyond traditional risk factors or measures of atherosclerosis alone.

Previous studies have often assessed BAC as a binary variable (presence vs absence)[13] or, more recently, using semi-quantitative or relative scoring systems.[14,15] Our own prior work moved towards quantification using patch-based segmentation models, which had adequate sensitivity to detect disease progression[16,17]; however, these models suffered from false positives due to imaging artefacts in breast tissue. To address this, we developed the more advanced transformer-based segmentation model used in this study, which maintains high sensitivity while reducing false-positive detections related to artefacts or nonvascular calcifications. This new model enables a standardized, automated quantification of BAC as an absolute physical metric (in mm^2), an approach analogous to the Agatston score that has been a long-standing goal in the field but has lacked large-scale validation.[18] In this study, we evaluate whether this automated mm^2 quantification of BAC from routine screening mammograms enhances CVD prediction in a large, multiracial cohort across two geographically diverse regions. This automated strategy has the potential to provide opportunistic cardiovascular risk assessment in women already undergoing routine screening mammography.

Methods

Study population

Our study included 74 124 women from Emory Healthcare (Atlanta, GA, USA) in the Emory Breast Imaging Dataset (EMBED)[19] who served as our internal cohort for development and validation and 49 638 women from Mayo Clinic Enterprise (Arizona, Florida, Mid-West, Rochester) who constituted our external validation cohort (see Supplementary data online, *Figure S1*). Inclusion criteria included age 40–79 years and any face-to-face clinical follow-up following an initial screening mammogram. For each patient from Emory Healthcare, the earliest screening mammogram was included for evaluation to maximize the follow-up period. At Mayo Clinic, all mammogram samples were from 2018. For all analyses, only a single index mammogram was used per patient.

In order to evaluate BAC as an opportunistic screening risk marker, we excluded women with prior known CVD, women who had existing records of major adverse cardiovascular events (MACE), and women who had CAC scoring or coronary angiography before the index mammogram. These strict inclusion criteria maximize the likelihood of excluding any women who had extant CVD that could artificially inflate the demonstrated prognostic value of BAC for asymptomatic screening.

Measurement of breast arterial calcification

Artificial intelligence model development

To increase the BAC segmentation accuracy over the previously published patch-based segmentation model[16] that does not fully leverage the spatial location of the patches, we developed a proprietary transformer-based neural network architecture (see Supplementary data online, *Figure S2*) to directly segment and quantify BAC within the mediolateral oblique views of high resolution, two-dimensional, full-field digital screening mammograms. Mediolateral oblique images are used since this plane is more perpendicular to breast arteries compared with the craniocaudal view. To capture hierarchical information, we created a set of Mix Transformer encoders that use the same design but come in different sizes. Inspired by the structure and attention method used in Pyramid vision transformer network,[20] we added new features like overlapping patch merging and removing positional encoding. We applied a decoder made only of multilayer perceptron layers, avoiding the complex and handcrafted components. This simple design is effective for BAC segmentation because our hierarchical transformer encoder has a much larger effective receptive field compared with traditional convolutional neural network encoders. The model was trained, validated, and tested using a set of 1000 screening mammograms annotated by an expert radiologist (HT) and divided into 90%–5%–5% training, validation, and test exams.

The final model output is a binary mask with detected BAC pixels. Post-processing is applied to differentiate between calcified pixels in a vessel and background non-calcified vessel. Breast arterial calcification area is quantified by computing the number of segmented pixels multiplied by the pixel size defined in the DICOM metadata to yield an area in mm^2. Model performance compared with expert annotation for area quantification resulted in a mean absolute error of 3.94 mm^2, R^2 of 0.91, and Pearson correlation of 0.95, which supersedes previously reported inter-observer variability for BAC which ranges from 0.48 to 0.53.[8]

Breast arterial calcification severity classification

In this study, both left and right breasts were included in the quantification, and the maximum BAC area between the two breasts was used as the final BAC area for a patient. Breast arterial calcification severity was categorized into four groups based on the quantified BAC area: zero BAC (0 mm^2), mild (>0–10 mm^2), moderate (>10–25 mm^2), and severe (>25 mm^2) (*Figure 1*). These thresholds were established through empirical analysis that maximized between-group hazard ratio (HR) differences in the Emory cohort (see Supplementary data online, *Figure S3*).

The zero BAC was validated through visual inspection of an independent set of 500 exams (see Supplementary data online, *Table S1*). Bland–Altman analysis was performed to evaluate the agreement between the artificial intelligence (AI)–quantified BAC and the ground truth scores in this set of 500 exams. Further

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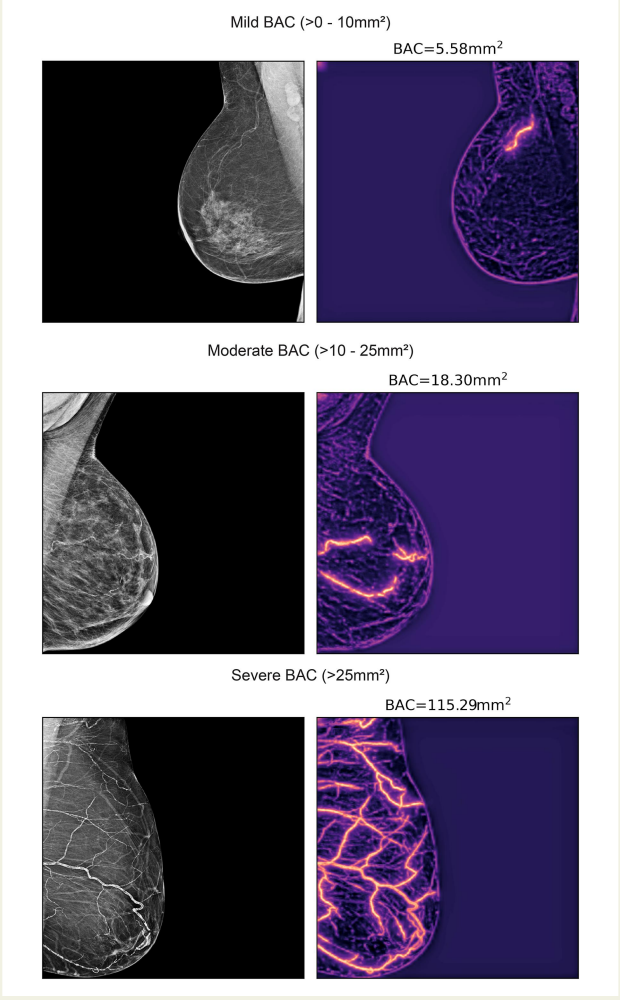


Figure 1 Examples of mammograms with mild, moderate, and severe breast arterial calcification quantified by our artificial intelligence model. *Left*: Original image. *Right*: Artificial intelligence model heat map with breast arterial calcification score shown above

qualitative analysis was done to evaluate true positive BAC in the presence of non-vascular calcifications, markers, and breast implants, as well as false positives.

Identification of cardiovascular events, risk factors, and PREVENT score

Using a previous expert-curated list, the primary/principal diagnostic ICD codes were extracted from the electronic health record (EHR) with timestamp data from both clinical sites to identify MACE[21] (see Supplementary data online, *Table S2*) and all-cause mortality. For women without events at Emory, a face-to-face visit following the mammogram date was required, defined by an in-person visit to a healthcare provider at the institution.

To compare BAC to the PREVENT risk score,[5] risk factors including total and HDL cholesterol, systolic blood pressure, body mass index, estimated glomerular filtration rate (eGFR), diabetes (see Supplementary data online, *Table S3*), smoking, use of antihypertensive medication, and use of statins were extracted from the EHR as ICD codes or laboratory values. PREVENT scores were calculated by implementing the algorithm from Khan *et al.*[5] We validated our implementation by manually comparing computed scores of 40 random patients against the American Heart Association PREVENT online calculator (see Supplementary data online, *Figure S4*).[22] The PREVENT score calculator was then applied to all eligible Emory and Mayo patients with complete risk factor data.

Statistical analysis

Baseline characteristics

We compared the prevalence of various demographics and risk factors between BAC severity groups and presence or absence of MACE using several statistical tests: χ^2 for discrete variables and *t*-tests or one-way ANOVA for continuous variables. The Kruskal–Wallis test was used to assess the differences in median follow-up duration between groups.

Primary and secondary time-to-event analyses

For all time-to-event analyses, the follow-up period for each patient began at the date of their index mammogram. Patients were followed until the first occurrence of a MACE. Censoring occurred at the date of last known clinical follow-up or at the administrative study cut-off date of 1 September 2024.

The association between BAC severity and MACE incidence was examined using Kaplan–Meier curves, with differences assessed by the log-rank test. Our primary analysis utilized this time-from-study-entry as the timescale. A secondary analysis was conducted using age as the primary timescale, consistent with methodologies such as PREVENT, to account for strong confounding effect of age on CVD and to adjust for left-truncation due to varying patient ages at entry into the study.

Subgroup analyses

To investigate potential differential effects, subgroup analyses were performed. First, given the potential prognostic value in younger populations, a dedicated analysis was conducted in patients under 50 years of age, specifically examining MACE outcomes in those with moderate to severe BAC compared with those without BAC. Second, to align with established risk stratification, additional subgroup analyses were conducted within different CVD risk categories (low, borderline, intermediate, and high) as defined by PREVENT. Within these risk categories, event-free survival was also compared between moderate to severe BAC and zero BAC patients. Differences in event-free survival rates between BAC severity groups were assessed using the log-rank test.

Predictive model development and specification

To assess the incremental predictive value of BAC, we developed and compared three distinct Cox proportional hazards models:

(1) Age + BAC model: included only age and BAC
(2) PREVENT-only model: utilized the PREVENT risk score as the predictor
(3) PREVENT + BAC model: combined PREVENT risk score with BAC

In models incorporating BAC (Models 1 and 3), we examined BAC separately as a categorical (zero, mild, moderate, severe) and a continuous variable. For completeness, we explored non-linear associations by transforming BAC using $\log_2(\text{BAC}+1)$ as was done in studies of coronary artery calcium,[23,24] allowing us to examine how a doubling of the BAC measurement affects event risk. Adding 1 to BAC before transformation enables us to include zero BAC values in our analysis.

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Inferential modelling and competing risk analysis

The Cox proportional hazards models were fitted on the entire study cohort to obtain stable HRs and 95% confidence intervals (CIs) for inferential purposes. The proportional hazards assumption was verified using Schoenfeld residuals, with no significant violations detected. Furthermore, to analyse the specific non-fatal outcomes of acute myocardial infarction (AMI), stroke, and heart failure while accounting for the competing risk of mortality, we fitted Fine–Gray subdistribution hazards models and report the resulting subdistribution HRs (sHRs) and their 95% CIs.

Model validation and performance comparison

For rigorous validation, the analysis was restricted to the subset of the cohort for which PREVENT data were available to ensure a fair comparison across all models. This subset was then randomly partitioned into training (50%), validation (20%), and held-out test (20%) sets. All models were developed on the training set. The validation set was used as a checkpoint to assess the need for model recalibration. The criterion was that no recalibration would be performed if the model demonstrated good calibration, defined by a 95% CI for the calibration slope containing 1.0 and for the intercept containing 0.0. Based on this assessment, the models were found to be well-calibrated, and no changes were made before final evaluation on the test set. Model performance was ultimately assessed on the test set for discrimination using the concordance index (C-index) and for calibration using the calibration intercept and slope.

To statistically compare model performance, a non-parametric bootstrap procedure with 1000 iterations was employed on the test set. The pairwise difference in the C-index between models was calculated for each iteration to generate a mean difference and a 95% CI using the percentile method. The Bonferroni correction was applied to account for multiple comparisons, resulting in a significance threshold of $P < .0167$. A similar bootstrap procedure was used to generate 95% CIs for calibration metrics.

Analyses were performed in Python (version 3.12) using the packages Lifelines (version 0.30.0). The subdistribution analysis of competing risks was done using the cmprsk package (version 2.2-12) in R (version 4.5.0).

Results

Study characteristics

The study included an internal (Emory) and an external (Mayo Clinic) cohort of women with a mean age of 55.5 [standard deviation (SD): 10.4] and 59.5 (SD: 10.2) years, respectively (*Table 1*). The median follow-up period was 7 years [interquartile range (IQR): 4–10]. Breast arterial calcification was detected in 16.1% of women in the internal cohort and 26.0% in the external cohort.

Breast arterial calcification severity was strongly associated with the incidence of MACE and all-cause mortality. A pronounced dose–response relationship was observed in both cohorts, where increasing BAC severity corresponded to a stepwise increase in event rates (see Supplementary data online, *Table S4*). In the internal cohort, the incidence of MACE increased more than eight-fold, from 5.96 per 1000 person-years in women with zero BAC to 48.89 in those with severe BAC. This risk gradient was evident for all individual outcomes, including AMI, stroke, heart failure, and death. Notably, event rates were consistently higher in the external cohort compared with the internal cohort across all levels of BAC severity.

The severity of BAC was also strongly correlated with patient age and established cardiometabolic risk factors. There was a mean age difference of approximately 13 years between patients with zero BAC and those with severe BAC ($P < .001$) (*Table 1*). Furthermore, factors including diabetes, use of antihypertensive medications and statins, higher systolic blood pressure, higher body mass index, and lower eGFR were all positively associated with increasing BAC severity. In contrast, smoking was not found to be associated with BAC in either cohort.

Breast arterial calcification as an independent risk factor for major adverse cardiovascular events

Kaplan–Meier analysis of the combined cohorts revealed distinct graded separation between zero, mild, moderate, and severe BAC for all outcomes (AMI, stroke, heart failure, and all-cause mortality) (*Figure 2*). Corresponding analysis using age as the timescale is provided in Supplementary data online, *Figure S5*. Notably, in patients younger than 50 years (*Figure 3* and Supplementary data online, *Figure S6*), where BAC is less frequently observed, moderate to severe BAC was associated with significantly lower event-free survival for MACE when compared with zero BAC.

In multivariable Cox analysis adjusting for age, BAC severity demonstrated a clear, graded dose–response relationship with risk for all adverse outcomes (*Table 2*). While hazards were elevated across all categories, the association was strongest and most consistent for the severe BAC group. The most pronounced effects were seen in heart failure for the internal cohort (severe BAC: HR 2.13, 95% CI 1.67–2.73, $P < .001$) and in all-cause mortality for the external cohort (severe BAC: HR 2.00, 95% CI 1.63–2.45, $P < .001$). Age also remained a significant independent predictor across all models.

Finally, Fine–Gray analysis was performed to account for death as a competing event for each individual non-fatal outcome (AMI, stroke, heart failure) and their composite (see Supplementary data online, *Tables S5* and *S6*). The sHRs from the Fine–Gray models were of a greater magnitude than the corresponding HRs from the standard Cox models. For instance, the sHR for heart failure in the severe BAC group of the external cohort was 4.65 (95% CI 3.07–7.04, $P < .001$) (see Supplementary data online, *Table S6*), significantly higher than the Cox HR of 1.78 (95% CI 1.46–2.17, $P < .001$) (*Table 2*).

Analyses of BAC as a continuous variable further corroborated these findings. Each 1 mm^2 increase in BAC area was associated with a 1%–2% increased risk across all outcomes ($P < .001$) (*Table 2*). Similarly, a non-linear analysis showed that each doubling of the BAC area (a 1-unit increase in $\log_2[BAC + 1]$) increased the risk of any MACE by 7%–19% (see Supplementary data online, *Table S7*).

Incremental prognostic value of breast arterial calcification over PREVENT scores

The PREVENT risk score was available for a subgroup of 4876 (6.6%) women in the internal cohort and 13 564 (27.9%) in the external cohort (see Supplementary data online, *Table S8*). This subgroup was older [internal cohort: mean age 59.1 years (SD: 10.0); external cohort: mean age 64.1 years (SD: 10)] and at

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Table 1 Baseline demographic characteristics and risk factors based on breast arterial calcification severity and whether a study participant had any subsequent major adverse cardiovascular events

	Overall	BAC severity					MACE		
		Zero	Mild	Moderate	Severe	*P*-value	No event	Event	*P*-value
Internal cohort	74 124	62 187 (83.9)	10 369 (14.0)	854 (1.2)	714 (1.0)		70 530 (95.2)	3594 (4.8)	
Age, years, mean (SD)	55.5 (10.4)	54.6 (10.0)	58.9 (10.9)	68.4 (8.6)	67.8 (9.1)	<.001	55.1 (10.3)	62.8 (9.8)	<.001
Race, *n* (%)									
Asian	4940 (6.7)	4172 (6.7)	688 (6.6)	52 (6.1)	28 (3.9)	<.001	4835 (6.9)	105 (2.9)	<.001
Black	31 971 (43.1)	27 153 (43.7)	4068 (39.2)	338 (39.6)	412 (57.7)		29 805 (42.3)	2166 (60.3)	
Other	1423 (1.9)	1173 (1.9)	223 (2.2)	15 (1.8)	12 (1.7)		1401 (2.0)	22 (0.6)	
Unknown	5747 (7.8)	4886 (7.9)	771 (7.4)	52 (6.1)	38 (5.3)		5688 (8.1)	59 (1.6)	
White	30 043 (40.5)	24 803 (39.9)	4619 (44.5)	397 (46.5)	224 (31.4)		28 801 (40.8)	1242 (34.6)	
Hispanic	4010 (5.4)	3311 (5.3)	602 (5.8)	57 (6.7)	40 (5.6)	.079	3917 (5.6)	93 (2.6)	<.001
Follow-up, years, median [Q1, Q3]	7.0 [4.0, 10.0]	7.0 [4.0, 10.0]	7.0 [4.0, 9.0]	6.0 [4.0, 9.0]	5.0 [3.0, 8.0]	<.001	7.0 [4.0, 10.0]	4.0 [2.0, 6.0]	<.001
Smoking, *n* (%)	3881 (22.4)	3145 (22.2)	589 (22.9)	78 (25.7)	69 (23.8)	.413	3318 (21.3)	563 (31.4)	<.001
Diabetes, *n* (%)	12 663 (17.1)	9951 (16.0)	2077 (20.0)	307 (35.9)	328 (45.9)	<.001	10 859 (15.4)	1804 (50.2)	<.001
Anti-HT medication, *n* (%)	4910 (6.6)	3723 (6.0)	858 (8.3)	147 (17.2)	182 (25.5)	<.001	3955 (5.6)	955 (26.6)	<.001
Statins, *n* (%)	3381 (4.6)	2453 (3.9)	629 (6.1)	132 (15.5)	167 (23.4)	<.001	2453 (3.5)	928 (25.8)	<.001
Total cholesterol, mg/dL, mean (SD)	193.3 (34.4)	193.5 (34.3)	193.2 (34.8)	189.6 (34.1)	187.4 (36.3)	.159	193.6 (34.1)	190.2 (36.8)	.01
HDL cholesterol, mg/dL, mean (SD)	59.2 (15.3)	59.2 (15.2)	59.6 (15.5)	60.6 (15.6)	56.7 (15.9)	.16	59.6 (15.2)	55.8 (15.3)	<.001
SBP, mmHg, mean (SD)	128.8 (18.8)	128.4 (18.6)	130.0 (19.3)	133.7 (19.9)	137.9 (21.3)	<.001	128.3 (18.5)	135.4 (21.5)	<.001
BMI, kg/m^2, mean (SD)	28.4 (5.4)	28.5 (5.4)	27.8 (5.5)	28.5 (5.3)	29.6 (5.4)	<.001	28.3 (5.4)	29.6 (5.6)	<.001
eGFR, mL/min/1.73 m^2, mean (SD)	83.5 (20.4)	84.8 (19.5)	80.6 (21.6)	68.6 (23.3)	60.7 (28.6)	<.001	84.7 (19.5)	72.6 (25.1)	<.001
External cohort	49 638	36 737 (74.0)	10 725 (21.6)	1582 (3.2)	604 (1.2)		43 622 (87.9)	6016 (12.1)	
Age, years, mean (SD)	59.5 (10.2)	58.5 (9.9)	63.3 (10.2)	70.4 (6.6)	71.1 (6.9)	<.001	58.5 (10.0)	66.4 (8.8)	<.001
Race, *n* (%)									
Asian	893 (1.8)	808 (2.2)	139 (1.3)	14 (0.9)	8 (1.4)	<.001	828 (1.9)	60 (1.0)	<.001
Black	546 (1.1)	330 (0.9)	128 (1.2)	17 (1.1)	9 (1.5)		479 (1.1)	60 (1.0)	
Other	943 (1.9)	734 (2.0)	203 (1.9)	25 (1.6)	10 (1.7)		872 (2.0)	54 (0.9)	
White	47 255 (95.2)	34 863 (94.9)	10 242 (95.5)	1525 (96.4)	576 (95.4)		41 440 (95.0)	5841 (97.1)	
Hispanic	878 (1.77)	750 (1.92)	175 (1.64)	19 (1.26)	10 (1.68)	<.001	802 (1.84)	67 (1.13)	<.001

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Table 1 *Continued*

	Overall	BAC severity					MACE		
		Zero	Mild	Moderate	Severe	*P*-value	No event	Event	*P*-value
Follow-up, years, median [Q1, Q3]	7.0 [6.0, 7.0]	7.0 [6.0, 7.0]	7.0 [6.0, 7.0]	6.0 [5.0,7.0]	6.0 [3.0, 7.0]		7.0 [6.0, 7.0]	3.0 [2.0, 5.0]	<.001
Smoking, *n* (%)	5629 (41.5)	4545 (40.6)	790 (41.4)	204 (42.4)	112 (45.5)	.42	3998 (40.0)	1771 (49.6)	<.001
Diabetes, %	14.44	17.08	42.23	72.15	68.98	<.001	10.86	34.41	<.01
Anti-HT medication, %	0.52	0.7	1.58	2.45	2.49	<.001	.32	2.19	<.001
Statins, %	2.82	3.36	8.94	14.59	16.73	<.001	1.44	14.47	<.001
Total cholesterol, mg/dL, mean (SD)	195.8 (43.7)	198.58 (43.11)	195.57 (44.2)	192.63 (43.74)	184.79 (43.63)	<.001	198 (42.99)	183.17 (44.96)	<.001
HDL cholesterol, mg/dL, mean (SD)	63.9 (19.77)	65.96 (20.44)	62.98 (19.29)	60.98 (18.22)	60.49 (18.62)	<.001	64.68 (19.72)	60.02 (19.56)	
SBP, mmHg, mean (SD)	131.3 (19.5)	130.2 (19.5)	133.2 (19.1)	134.5 (19.3)	136.4 (21.9)	<.001	131.4 (19.2)	136.4 (21.5)	
BMI, kg/m^2, mean (SD)	24.7 (9.8)	23.4 (10.4)	25.5 (9.8)	28.2 (10.5)	27.4 (10.8)	<.001	24.2 (10.2)	27.4 (10.8)	
eGFR, mL/min/1.73 m^2, mean (SD)	93.2 (20.4)	103.4 (19.3)	98.7 (19.2)	93.3 (19.5)	84.5 (19.1)	<.001	104.5 (19.5)	86.5 (20.1)	<.001

BAC, breast arterial calcification; BMI, body mass index; eGFR, estimated glomerular filtration rate; HDL, high-density lipoprotein; HT, hypertensive; MACE, major adverse cardiovascular events; SBP, systolic blood pressure; SD, standard deviation.

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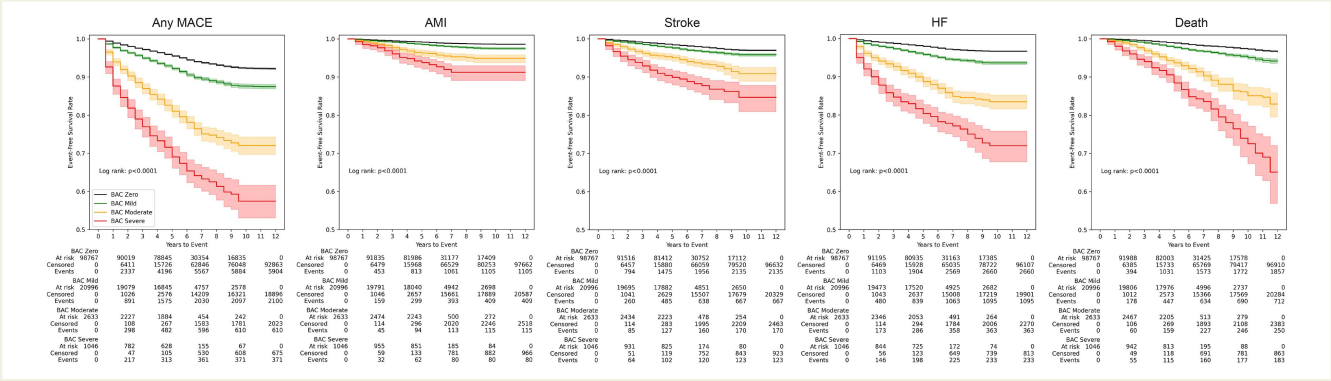


Figure 2 Kaplan–Meier estimates of event-free survival stratified by breast arterial calcification severity (zero, mild, moderate, severe). The tables below the plots indicate the number of patients at risk for each breast arterial calcification severity group at 2-year intervals. Differences in event-free survival rates among the breast arterial calcification severity groups were assessed using the log-rank test. AMI, acute myocardial infarction; HF, heart failure; MACE, major adverse cardiovascular event

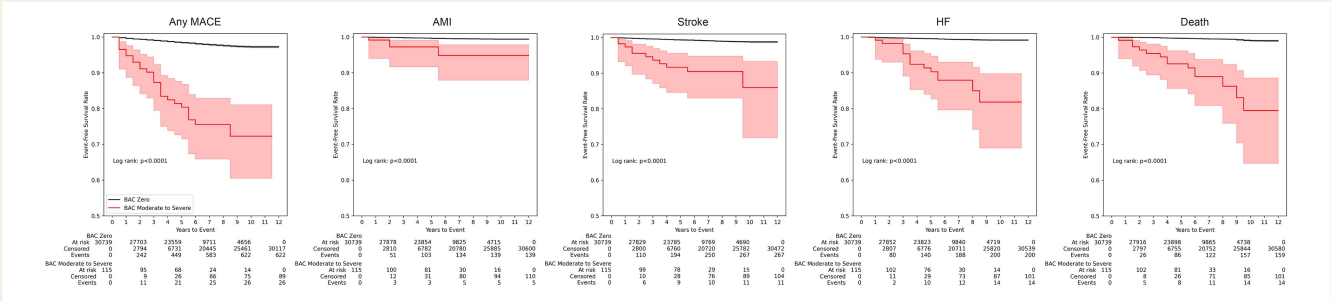


Figure 3 Kaplan–Meier estimates of event-free survival in patients under 50 years of age, stratified by zero breast arterial calcification and moderate to severe breast arterial calcification. Years to event represents the time since the initial mammogram to the occurrence of a major adverse cardiovascular event or last follow-up. AMI, acute myocardial infarction; HF, heart failure

higher baseline risk, with a MACE incidence of 9.9% in the internal and 26.3% in the external cohort.

A direct comparison of unadjusted incidence rates showed that BAC severity was associated with a wider range of MACE rates compared with the PREVENT risk categories (see Supplementary data online, *Table S4*). In the external cohort, the MACE rate increased over four-fold from 35.95 per 1000 person-years in the zero BAC group to 154.35 in the severe group. Notably, the event rate for patients in the severe BAC category exceeded the rate for those in the high-risk PREVENT category in both cohorts.

These trends were consistent with survival analyses. Kaplan–Meier curves demonstrated that moderate to severe BAC was associated with significantly lower MACE-free survival, even within the same PREVENT risk category (*Figure 4*; Supplementary data online, *Figure S7*). After adjusting for PREVENT risk scores in multivariable Cox models, BAC remained a significant and independent predictor of MACE in a dose-dependent manner (*Table 3*). In the internal cohort, HRs for any MACE were 1.32 (95% CI 1.10–1.59) for mild, 1.75 (95% CI 1.23–2.50) for moderate, and 3.29 (95% CI 2.15–5.05) for severe BAC. The corresponding HRs in the external cohort were 1.28 (95% CI 1.17–1.39), 1.79 (95% CI 1.55–2.06), and 2.80 (95% CI 2.36–3.32), respectively. Further analysis of specific outcomes revealed that while BAC was not a significant predictor for AMI in the internal cohort after adjusting for PREVENT, it remained significantly associated with AMI in the external cohort, with HRs of 1.42 (95% CI 1.08–1.86) for moderate and 2.47 (95% CI 1.79–3.40) for severe BAC.

In the competing risk analysis accounting death as a competing event, BAC severity showed its strongest association with incident heart failure in the internal cohort (severe BAC: sHR 3.89, 95% CI 2.05–7.39, $P < .001$) (see Supplementary data online, *Table S5*) and AMI in the external cohort (severe BAC: sHR 3.28, 95% CI 2.32–4.64, $P < .001$) (see Supplementary data online, *Table S6*). In the same analysis, the PREVENT score was also a robust predictor for all three non-fatal outcomes.

Adding BAC as a categorical variable to the PREVENT model significantly improved risk discrimination in both cohorts (see Supplementary data online, *Figure S8*). In the internal cohort, the C-index increased from 0.71 for the PREVENT-only model to 0.73 (see Supplementary data online, *Table S9*) for the PREVENT + BAC model (mean difference: 0.02, 95% CI: 0.004–0.032, $P < .001$) (see Supplementary data online, *Figure S8*). Similarly, in the external cohort, the C-index increased from 0.62 for the PREVENT-only model to 0.64 (see Supplementary data online, *Table S9*) for the PREVENT + BAC model (mean difference: 0.02, 95% CI 0.012–0.029, $P < .001$) (see Supplementary data online, *Figure S8*). The enhanced model also remained well-calibrated (see

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Table 2 Association of breast arterial calcification and clinical outcomes in the age + breast arterial calcification model

Model	Covariate		Any MACE			AMI			Stroke			HF			Death		
			n (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value
Internal cohort (Emory)																	
BAC categorical model	BAC	Zero (*n* = 62 187)	2599 (4.18)	Reference		422 (0.68)	Reference		1258 (2.02)	Reference		595 (0.96)	Reference		797 (1.28)	Reference	
		Mild (*n* = 10 369)	683 (6.59)	1.087 (1.027–1.150)	.004	112 (1.08)	1.029 (0.966–1.096)	.377	300 (2.89)	1.044 (0.983–1.109)	.159	183 (1.76)	1.054 (0.991–1.122)	.096	240 (2.31)	1.067 (1.004–1.135)	.037
		Moderate (*n* = 1097)	180 (16.41)	1.528 (1.312–1.779)	<.001	29 (2.64)	1.171 (0.978–1.401)	.086	83 (7.57)	1.366 (1.158–1.611)	<.001	48 (4.38)	1.278 (1.073–1.522)	.006	81 (7.38)	1.479 (1.250–1.750)	<.001
		Severe (*n* = 471)	132 (28.03)	2.165 (1.733–2.706)	<.001	20 (4.25)	1.342 (1.023–1.760)	.034	57 (12.10)	1.742 (1.366–2.221)	<.001	58 (12.31)	2.128 (1.658–2.731)	<.001	55 (11.68)	1.926 (1.503–2.468)	<.001
	Age	Per 1-year increase		1.019 (1.017–1.021)	<.001		1.004 (1.002–1.007)	<.001		1.011 (1.009–1.013)	<.001		1.008 (1.006–1.010)	<.001		1.008 (1.006–1.010)	<.001
	Interaction	Age × mild BAC		1.002 (1.001–1.003)	<.001		1.001 (1.000–1.002)	.177		1.001 (1.000–1.002)	.025		1.001 (1.000–1.002)	.020		1.001 (1.000–1.002)	.005
		Age × moderate BAC		1.006 (1.004–1.008)	<.001		1.002 (1.000–1.005)	.085		1.004 (1.002–1.007)	<.001		1.003 (1.001–1.006)	.008		1.006 (1.003–1.008)	<.001
		Age × severe BAC		1.010 (1.007–1.014)	<.001		1.004 (1.000–1.008)	.048		1.008 (1.004–1.011)	<.001		1.011 (1.007–1.015)	<.001		1.009 (1.006–1.013)	<.001
BAC linear model	BAC	Per 1 mm^2 increase		1.020 (1.014–1.026)	<.001		1.012 (1.005–1.019)	<.001		1.017 (1.011–1.023)	<.001		1.021 (1.014–1.027)	<.001		1.020 (1.014–1.027)	<.001
	Age	Per 1-year increase		1.029 (1.027–1.031)	<.001		1.008 (1.005–1.011)	<.001		1.018 (1.016–1.021)	<.001		1.013 (1.011–1.016)	<.001		1.013 (1.010–1.016)	<.001
	Interaction	Age × BAC		1.0002 (1.0001–1.0003)	<.001		1.0002 (1.0001–1.0003)	.001		1.0002 (1.0001–1.0003)	<.001		1.0003 (1.0002–1.0004)	<.001		1.0003 (1.0002–1.0004)	<.001
External cohort (Mayo Clinic)																	
BAC categorical model	BAC	Zero (*n* = 36 727)	3738 (10.18)	Reference		683 (1.86)	Reference		877 (2.39)	Reference		2065 (5.62)	Reference		1060 (2.89)	Reference	
		Mild (*n* = 10 725)	1547 (14.42)	1.045 (0.990–1.103)	.110	297 (2.77)	1.042 (0.979–1.108)	.197	367 (3.42)	1.045 (0.983–1.111)	.155	912 (8.50)	1.060 (1.001–1.122)	.045	472 (4.40)	1.055 (0.994–1.120)	.079
		Moderate (*n* = 1582)	470 (29.71)	1.385 (1.227–1.564)	<.001	86 (5.44)	1.223 (1.061–1.409)	.005	87 (5.50)	1.163 (1.011–1.338)	.034	315 (19.91)	1.449 (1.279–1.643)	<.001	169 (10.68)	1.392 (1.218–1.591)	<.001
		Severe (*n* = 604)	261 (43.21)	1.762 (1.456–2.131)	<.001	60 (9.93)	1.553 (1.248–1.931)	<.001	66 (10.93)	1.533 (1.236–1.901)	<.001	175 (28.97)	1.782 (1.465–2.167)	<.001	128 (21.19)	2.001 (1.634–2.450)	<.001
	Age	Per 1-year increase		1.037 (1.035–1.039)	<.001		1.011 (1.008–1.013)	<.001		1.013 (1.010–1.015)	<.001		1.030 (1.028–1.032)	<.001		1.017 (1.015–1.020)	<.001
	Interaction	Age × mild BAC		1.002 (1.001–1.003)	<0.001		1.001 (1.000–1.002)	0.029		1.001 (1.000–1.002)	0.011		1.002 (1.001–1.003)	<0.001		1.001 (1.001–1.002)	0.003
					<.001			.004			.022			<.001			<.001

Continued

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Table 2 *Continued*

Model	Covariate		Any MACE		AMI		Stroke		HF		Death	
		n (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value
	Age × moderate BAC		1.005 (1.003–1.006)			1.003 (1.001–1.005)			1.002 (1.000–1.004)			1.005 (1.004–1.007)	
	Age × severe BAC		1.007 (1.005–1.010)	<.001		1.006 (1.003–1.009)	<.001		1.006 (1.003–1.009)	<.001		1.008 (1.006–1.011)	<.001
BAC linear model	BAC	Per 1 mm^2 increase	1.015 (1.011–1.019)	<.001		1.013 (1.008–1.017)	<.001		1.011 (1.007–1.016)	<.001		1.015 (1.011–1.019)	<.001
	Age	Per 1-year increase	1.037 (1.035–1.039)	<.001		1.010 (1.008–1.013)	<.001		1.013 (1.010–1.015)	<.001		1.031 (1.028–1.033)	<.001
	Interaction	Age × BAC	1.0002 (1.0001–1.0003)	<.001		1.0002 (1.0001–1.0003)	<.001		1.0002 (1.0001–1.0003)	<.001		1.0002 (1.0001–1.0003)	<.001

The total number of cases of each event for BAC and PREVENT subgroups is presented as *n* (%). Interactions between age × BAC for the linear models are positive, but miniscule, demonstrating that the synergy of age and BAC is no more informative to prediction than considering age and BAC as independent covariates.
AMI, acute myocardial infarction; CI, confidence interval; HF, heart failure; HR, hazard ratio; MACE, major adverse cardiovascular event.

Supplementary data online, *Table S10* and *Figure S9*), with slopes of 1.041 and 1.120 for the internal and external cohorts, respectively.

Finally, when analysed as a continuous variable adjusted for PREVENT scores, a 1 mm^2 increase in BAC was associated with a 2%–3% increased risk across all outcomes, and each doubling of BAC was associated with an 11%–27% increased risk across all outcomes (*Table 3*; Supplementary data online, *Table S7*). Adding continuous BAC to the PREVENT model did not yield a statistically significant improvement in the C-index ($P = .32$) for the internal cohort; however, significant improvement was seen in the external cohort from a C-index of 0.62–0.64 (mean difference: 0.02; 95% CI 0.008–0.022, $P < .001$).

Artificial intelligence model robustness and generalizability

Our deep learning model for automated BAC quantification demonstrated robustness in the independent set of 500 exams as well as across a multi-site screening mammography population. In the set of 500 exams, for the detection of BAC, the model's sensitivity was 0.91 and specificity was 0.95 (see Supplementary data online, *Figure S10*). Bland–Altman analysis showed a low systematic bias with a mean bias of −0.0455 (95% limits of agreement: −5.14 to +5.05) (see Supplementary data online, *Figure S11*). Our model was robust to the presence of nonvascular calcifications, breast implant, and common mammographic artefacts, including mole markers (see Supplementary data online, *Figure S12*). Infrequent false positives were observed for very dense ductal calcifications (see Supplementary data online, *Figure S13*), but these instances yielded very low predicted scores (0.88 mm^2, IQR: 0.15–2.44 mm^2) and were distinct from true BAC (see Supplementary data online, *Figure S10*). Finally, the model's performance was also consistent across Hologic, GE, and LORAD scanners (see Supplementary data online, *Tables S1* and *S11*) and the prognostic value of the BAC score remained significant after stratification by breast density, with Kaplan–Meier curves showing significant separation across BAC severity levels in the main multi-site screening mammography population (see Supplementary data online, *Figure S14*).

Discussion

Our study demonstrates that an automated, AI-driven quantification of BAC on routine screening mammograms is a strong, independent predictor of adverse cardiovascular events in a large, multiracial population. We show that this measure not only stratifies risk across a spectrum or severity but also offers significant prognostic value beyond the recently recommended PREVENT score,[5,6] highlighting its potential as a widespread, opportunistic screening tool for women (*Structured Graphical Abstract*).

Our quantitative deep learning approach helps advance the field towards an 'Agatston score for mammography',[18] moving beyond the moderate inter-observer agreement of traditional qualitative scoring.[8] To further this goal, our approach introduces a key innovation by quantifying BAC as an absolute, physical metric (mm^2). This provides an interpretable and reproducible measurement, analogous to the Agatston score itself, which can be standardized for both clinical and research applications, in contrast to other recently developed relative scoring systems.[14,15]

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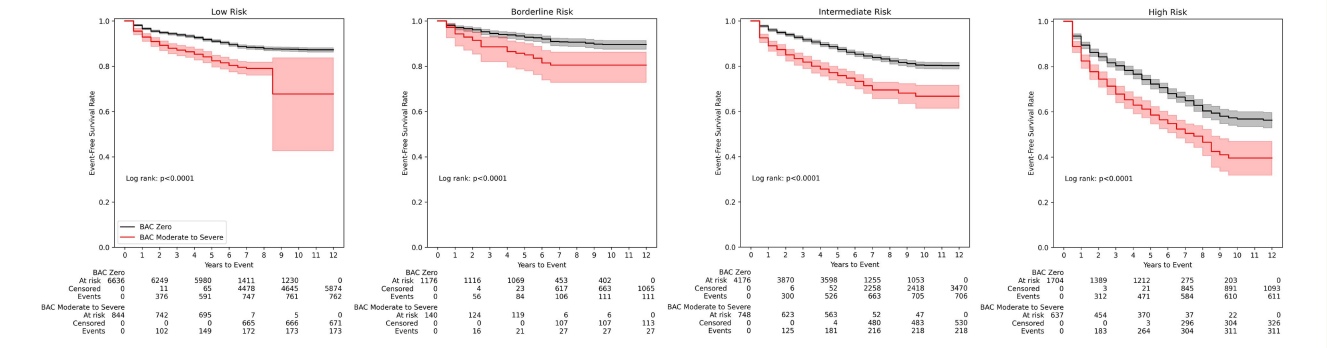


Figure 4 Kaplan–Meier estimates of event-free survival in low, borderline, intermediate, and high-risk patients identified by PREVENT, stratified by zero breast arterial calcification and moderate to severe breast arterial calcification

This quantitative approach enables a more precise analysis of the relationship between clinical risk factors and BAC severity. In our study, we observed that as BAC severity increased, there were significantly higher proportions of patients with diabetes mellitus and use of antihypertensive medications and statins. Higher systolic blood pressure, higher body mass index, and lower eGFR also showed a positive correlation with BAC severity. These findings align with previous studies that reported higher prevalence of these risk factors in BAC-positive patients.[25,26] Notably, and in contrast to some reports[8,25] but consistent with others,[27] we found no association between BAC and smoking prevalence in either cohort.

A key finding is the incremental prognostic value of BAC when added to the established PREVENT model. In multivariable models adjusted for PREVENT, BAC remained an independent predictor of MACE with clear dose–response relationship (e.g. internal cohort severe BAC: HR 3.29, 95% CI 2.15–5.05; external cohort severe BAC: HR 2.80, 95% CI 2.36–3.32). Adding BAC also improved risk discrimination in both cohorts (internal C-index: 0.71 to 0.73, $P < .001$; external C-index: 0.62 to 0.64, $P < .001$). Moreover, the prognostic strength of the continuous BAC (mm^2) metric itself was robust; when analysed as a continuous variable, both the linear (per 1 mm^2 increase) and the log-transformed (doubling) BAC area were significant predictors for nearly all MACE outcomes in both cohorts, with the only exception being AMI in the internal cohort. Furthermore, the enhanced PREVENT + BAC model remained well-calibrated in both cohorts (slopes: 1.041 and 1.120, respectively), suggesting generalizability of the findings.[28] This prognostic value was also confirmed using competing risk analysis (see Supplementary data online, *Tables S5* and *S6*), which adjusts for non-fatal MACE risk by accounting for all-cause mortality.[29,30] While this analysis showed a stronger association in the age-adjusted model, the PREVENT + BAC model remained prognostic in the external cohort, demonstrating robustness despite the internal cohort's lower event rate.

Beyond improving the overall model, we demonstrate that BAC can provide risk stratification within the established PREVENT categories. Kaplan–Meier analyses revealed that BAC's utility extends across the risk spectrum, though its impact varied between our cohorts. In the internal cohort, BAC was most prognostic for patients already classified as the intermediate or high risk. In the external cohort, however, BAC was prognostic across all PREVENT risk categories. This difference may be related to the external cohort's composition at a quaternary referral centre with an older patient cohort and substantially higher baseline MACE rate. This higher-risk profile, combined with recent evidence that PREVENT can demonstrate heterogeneous performance, with variable calibration and discrimination when transported to different US healthcare systems and populations,[31] may explain both the poor baseline performance of PREVENT in this group (C-index 0.62) and the broader prognostic utility of BAC.

This finding highlights the potential clinical utility of BAC, particularly in identifying risk among women not captured by traditional screening. In our study, the data required to calculate a PREVENT score was available for only 6.6% of the internal cohort and 27.9% of the external cohort. This reflects a real-world challenge in acquiring complete data for risk prediction. In contrast, screening mammography is performed on approximately 40 million women annually in the USA,[32] offering a readily available, opportunistic method to identify at-risk women without requiring additional tests. It is important to note that BAC quantification is not intended to replace comprehensive risk models like PREVENT. Rather, BAC may function as a powerful opportunistic identifier of at-risk patients who may otherwise be overlooked and drive formal cardiovascular risk assessment by their primary care physician.

The prognostic value of BAC is also especially noteworthy in women under 50. While BAC is more prevalent with age, its presence in younger women is a strong indicator of elevated cardiovascular risk.[14,33,34] Our analysis confirmed that moderate to severe BAC in this younger cohort was associated with significantly lower MACE-free survival, identifying a high-risk group that might otherwise be overlooked by age-weighted risk models.

The prevalence of BAC identified by our model was 16.1% in the internal cohort and 26.0% in the external cohort. The higher prevalence in the external cohort is likely attributable to the older average age of its patients, but both values are consistent with the 12.3%–29.4% prevalence range reported in the existing literature.[14,25,35,36] Our findings are also consistent with a growing body of literature establishing the prognostic value of BAC for CVD,[8,14,25,27,35] including a recent meta-analysis by Koh *et al.*[9] That analysis, however, did not find a statistically significant association between BAC and myocardial infarction.

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Table 3 Association of breast arterial calcification and clinical outcomes in the PREVENT + breast arterial calcification model

Model		Variable	Any MACE			AMI			Stroke			HF			Death		
			n (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value
Internal cohort (Emory)																	
BAC categorical model	BAC	Zero (*n* = 4008)	384 (9.58)	Reference		68 (1.70)	Reference		198 (4.94)	Reference		104 (2.59)	Reference		797 (1.28)	Reference	
		Mild (*n* = 711)	116 (16.32)	1.319 (1.098–1.585)	.003	28 (3.94)	1.313 (0.984–1.752)	.064	59 (8.30)	1.257 (0.998–1.584)	.053	36 (5.06)	1.236 (0.947–1.614)	.119	240 (2.31)	1.206 (0.924–1.574)	.168
		Moderate (112)	30 (26.79)	1.750 (1.225–2.501)	.002	5 (4.46)	1.305 (0.675–2.523)	.428	16 (14.29)	1.736 (1.087–2.773)	.021	14 (12.50)	2.185 (1.302–3.666)	.003	81 (7.38)	1.790 (1.034–3.097)	.037
		Severe (*n* = 45)	22 (48.89)	3.294 (2.150–5.047)	<.001	3 (6.67)	1.650 (0.624–4.367)	.313	8 (17.78)	2.056 (1.043–4.050)	.037	10 (22.22)	3.879 (1.997–7.537)	<.001	55 (11.68)	5.243 (2.880–9.544)	<.001
	PREVENT	Low (*n* = 1761)	57 (3.24)	Reference		13 (0.74)	Reference		29 (1.65)	Reference		12 (0.68)	Reference		14 (0.80)	Reference	
		Borderline (*n* = 600)	34 (5.67)	0.886 (0.685–1.148)	.361	7 (1.17)	0.894 (0.642–1.245)	.507	18 (3.00)	0.866 (0.646–1.159)	.332	9 (1.50)	0.856 (0.622–1.177)	.338	8 (1.33)	0.817 (0.595–1.123)	.214
		Intermediate (*n* = 1878)	268 (14.27)	1.788 (1.526–2.095)	<.001	42 (2.24)	1.072 (0.861–1.335)	.535	144 (7.67)	1.513 (1.259–1.819)	<.001	65 (3.46)	1.149 (0.934–1.413)	.190	86 (4.58)	1.322 (1.080–1.620)	.007
		High (*n* = 637)	194 (30.46)	3.688 (3.053–4.455)	<.001	42 (6.59)	1.947 (1.451–2.613)	<.001	90 (14.13)	2.432 (1.920–3.082)	<.001	78 (12.24)	2.822 (2.175–3.661)	<.001	65 (10.20)	2.273 (1.743–2.965)	<.001
BAC continuous model	BAC	BAC (per 1 mm^2 increase)		1.025 (1.016–1.034)	<.001		1.017 (0.997–1.036)	.091		1.019 (1.006–1.033)	.006		1.032 (1.019–1.046)	<.001		1.032 (1.018–1.046)	<.001
	PREVENT	PREVENT (per 1-point increase)		1.050 (1.044–1.056)	<.001		1.030 (1.019–1.041)	<.001		1.038 (1.030–1.046)	<.001		1.044 (1.035–1.053)	<.001		1.039 (1.030–1.049)	<.001
External cohort (Mayo Clinic)																	
BAC categorical model	BAC	Zero (*n* = 11 196)	2197 (19.62)	Reference		438 (3.91)	Reference		528 (4.72)	Reference		1372 (12.25)	Reference		609 (5.44)	Reference	
		Mild (*n* = 1910)	542 (28.38)	1.278 (1.170–1.395)	<.001	106 (5.55)	1.131 (0.969–1.320)	.118	128 (6.70)	1.163 (1.004–1.347)	.044	344 (18.01)	1.233 (1.109–1.370)	<.001	171 (8.95)	1.249 (1.090–1.432)	.001
		Moderate (*n* = 483)	200 (41.41)	1.788 (1.552–2.059)	<.001	39 (8.07)	1.421 (1.083–1.864)	.011	45 (9.32)	1.417 (1.095–1.835)	.008	130 (26.92)	1.680 (1.416–1.995)	<.001	81 (16.77)	1.975 (1.590–2.454)	<.001
		Severe (*n* = 248)	142 (57.26)	2.801 (2.362–3.320)	<.001	38 (15.32)	2.471 (1.794–3.401)	<.001	37 (14.92)	2.114 (1.542–2.899)	<.001	95 (38.31)	2.553 (2.077–3.139)	<.001	75 (30.24)	3.799 (2.982–4.840)	<.001
	PREVENT	Low (*n* = 6616)	1022 (15.45)	Reference		213 (3.22)	Reference		287 (4.34)	Reference		580 (8.77)	Reference		283 (4.28)	Reference	
		Borderline (*n* = 885)	134 (15.14)	0.902 (0.775–1.050)	.184	28 (3.16)	0.912 (0.722–1.151)	.437	22 (2.49)	0.768 (0.610–0.967)	.025	81 (9.15)	0.907 (0.758–1.086)	.288	32 (3.62)	0.830 (0.665–1.036)	.099
		Intermediate (*n* = 3969)	930 (23.43)	1.311 (1.215–1.415)	<.001	193 (4.86)	1.109 (0.982–1.253)	.097	202 (5.09)	1.016 (0.903–1.144)	.793	598 (15.07)	1.324 (1.210–1.450)	<.001	260 (6.55)	1.096 (0.979–1.227)	.111
		High (*n* = 2367)	995 (42.04)	2.402 (2.219–2.601)	<.001	187 (7.90)	1.477 (1.286–1.697)	<.001	227 (9.59)	1.538 (1.349–1.754)	<.001	682 (28.81)	2.434 (2.215–2.675)	<.001	361 (15.25)	2.076 (1.842–2.339)	<.001

Continued

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Table 3 *Continued*

Model	Variable		Any MACE			AMI			Stroke			HF			Death		
			n (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value	*n* (%)	HR (95% CI)	*P*-value
BAC continuous model	BAC	BAC (per 1 mm^2 increase)		1.018 (1.015–1.022)	<.001		1.016 (1.011–1.021)	<.001		1.015 (1.009–1.020)	<.001		1.018 (1.015–1.022)	<.001		1.026 (1.022–1.030)	<.001
	PREVENT	PREVENT (per 1-point increase)		1.027 (1.025–1.030)	<.001		1.011 (1.007–1.015)	<.001		1.010 (1.006–1.014)	<.001		1.029 (1.026–1.033)	<.001		1.022 (1.018–1.026)	<.001

The total number of cases of each event for BAC and PREVENT subgroups is presented as *n* (%).
AMI, acute myocardial infarction; CI, confidence interval; HF, heart failure; HR, hazard ratio; MACE, major adverse cardiovascular event.

Our results reflect this complexity, showing a significant association in our external cohort but a non-significant one in our internal cohort after adjusting for PREVENT.

Limitations

This study is conducted using retrospective data from two US institutions. We designed our study to validate the prognostic value of the BAC quantification across different populations, in contrast to developing a single, comprehensive risk model at one site that was externally validated at a second site. In other words, our goal was not to create a 'best fit' risk model that generalizes across populations, but rather to demonstrate the additive value of BAC as an independent predictor even in populations with differing baseline risk. The challenge of developing a universal risk model is reflected in data demonstrating that the atherosclerotic CVD and PREVENT risk calculators do not generalize well to certain populations.[6,31]

While our AI model's prognostic value was consistent across scanner manufacturers and breast densities, further work is needed to ensure performance across all scanners. Similarly, the population was approximately equally divided between Black and White patients for the internal dataset, but there were limited Asian, Hispanic, and Native American patients; therefore, generalizability to these populations requires further exploration. Major adverse cardiovascular event and risk data were extracted from the EHR using validated codes, but this could introduce bias from misclassification of events. Cardiovascular mortality is unreliable from EHR data, so all-cause mortality was used instead. Furthermore, data on important potential confounders such as menopause status, reproductive history, and social deprivation index were not reliably available in the EHR and therefore not included in the analyses.

While our findings demonstrate that BAC is significantly associated with adverse cardiovascular outcomes, we acknowledge that this retrospective observational study cannot establish causality. Rather, BAC should be considered a risk marker, not a causal risk factor. Whether BAC is a surrogate for unmeasured cardiovascular burden or reflects distinct vascular pathophysiology, such as medial arterial calcification, remains an important area for further study. Furthermore, because of missingness of data used to compute PREVENT scores, evaluation of BAC's additive value to prognostication with PREVENT scores was limited to a smaller subset of the population which could introduce selection bias; future work could evaluate this relationship after imputation of missing data.

We did not include metrics such as the net reclassification index, as this metric can be misleading in assessing the validity of new risk models.[37] While the C-index improvement was statistically significant, its modest magnitude, alongside the absence of a direct comparison to other established imaging markers like coronary artery calcium,[38] underscores that the incremental predictive value of BAC requires further prospective validation to determine its impact on clinical decision-making and cardiovascular outcomes.

Finally, while the current model functions on full-field digital mammography, a key area for future work will be the development of a BAC quantification model for digital breast tomosynthesis (DBT). As DBT is being increasingly used in the USA,[39,40] adapting this AI tool for DBT is an essential next

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step to maximize the opportunistic capture of high-risk patients at mammography.

Conclusion

Automated BAC quantification from routine mammography may provide an opportunistic and effective cardiovascular risk assessment method in women, without additional radiation exposure. Its predictive value is independent of traditional risk factors and PREVENT scores suggesting that BAC screening could enhance early cardiovascular risk detection in women undergoing routine mammography.

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Supplementary data

Supplementary data are available at *European Heart Journal* online.

Declarations

Disclosure of Interest

Nothing to declare.

Data Availability

A subset of the mammogram data from Emory used in this study is publicly available in the Emory Breast Imaging Dataset (EMBED) at https://doi.org/10.1148/ryai.220047 and can be accessed via https://registry.opendata.aws/emory-breast-imaging-dataset-embed/. The validation data from Mayo Clinic and the remaining Emory data cannot be shared publicly due to institutional restrictions. Limited information about these datasets may be shared on reasonable request to the corresponding author, subject to approval and appropriate confidentiality agreements.

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Ethical Approval

This study complied with all relevant ethical regulations. Study protocols received institutional review board approval at each study site, and a waiver of consent was granted due to the retrospective nature of this study. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines.

Pre-registered Clinical Trial Number

Not applicable.

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